



King Saud University
College of Computer and Information Sciences
Information Technology Department

IT222: Database Principles
2nd Semester 1447 H



THE RITZ-CARLTON

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Project Description:

This project is about building a database system to help manage The Ritz-Carlton hotel. The system is made to save and organize all the important information about the hotel's work, such as guest details, room bookings, staff tasks, and money payments. By putting all this data in one place, the system makes it much easier for the reception and the management to track guest stays and see the hotel's daily reports.

View Description:

This interface will be utilized by hotel personnel and administration to oversee and process booking information. It features comprehensive data regarding guest identities, room availability, order statuses, and transaction logs. The objective of this view is to provide the team with a transparent overview of occupancy levels, enable them to execute new

reservations seamlessly, and keep leadership updated on fiscal performance, ensuring a sophisticated and organized service standard.

Data Requirements:

- Customer

The Customer entity represents the individuals utilizing the hotel's services. Each user is uniquely identified by an IDNumber (Primary Key) to maintain high data integrity. This entity stores vital personal information such as the guest's full name (FullName), contact details (PhoneNumber, EmailAddress), and Nationality. A guest can secure multiple bookings over various periods, and the system retains a full history of their past stays.

- Room

The Room entity refers to the physical lodging units available for occupancy. Each unit is defined by a specific RoomNumber (Primary Key). The entity captures attributes such as the suite or room category (RoomType), floor location, guest capacity, and nightly rates (PricePerNight). The RoomStatus field allows staff to identify if a room is ready for check-in, currently occupied, or undergoing maintenance.

- Staff

The Staff entity signifies the workforce responsible for managing hotel functions. Each team member is designated a unique StaffID (Primary Key). The entity includes their full name, professional Role (e.g., Receptionist, Manager), and contact info. Staff members are tasked with overseeing room conditions and coordinating guest booking requests.

- Reservation

The Reservation entity represents the formal agreement between the guest and the hotel. It is identified using a composite primary key (IDNumber, RoomNumber) and includes arrival and departure dates, guest count, and the current state of the booking (e.g., confirmed, cancelled, or pending).

- Payment

The Payment entity focuses on fiscal transactions. Tracked by a TransactionRef (Primary Key), it monitors the amount paid, the date of the transaction, and the method used (e.g., credit card or cash) for each specific booking.

Transaction Requirements:

Data Entry:

1. Register Guest Profiles: Insert personal details and identification for new visitors into the database.
2. Submit New Bookings: Input stay specifications and select the appropriate room based on guest preferences.
3. Log Fiscal Transactions: Save billing records as soon as a guest completes the payment for their stay.

Data update/deletion:

1. Modify Room Status: Adjust the room's current state (e.g., switching to 'Occupied' during guest check-in).
2. Edit Reservation Details: Permit staff to change stay dates or update the booking status based on new developments.
3. Revise Staff Assignments: Update records regarding which employee is responsible for supervising specific rooms.

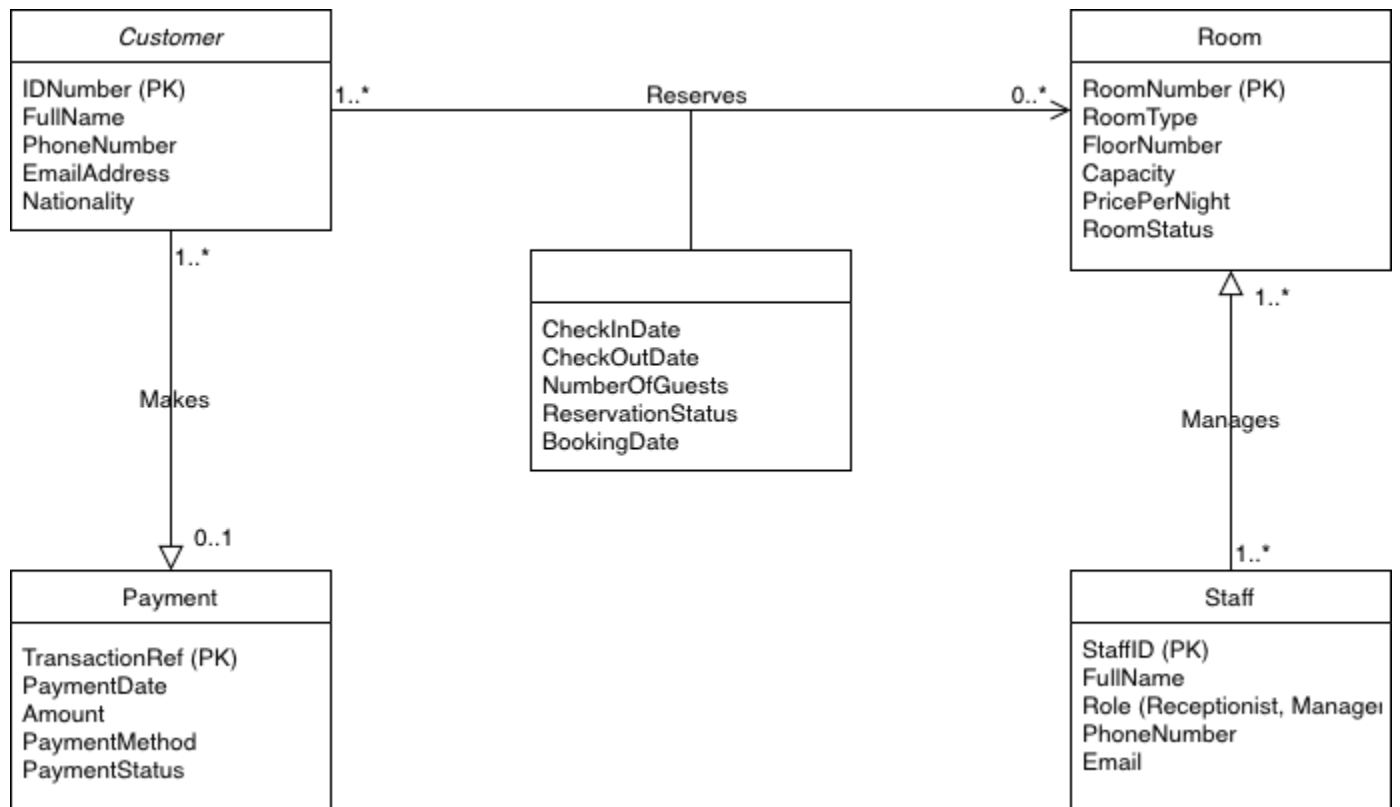
Data Queries:

1. Guest Profile
 - Description: Retrieves the fundamental identification details of a visitor.
 - Information: IDNumber, FullName, Nationality.
 - Source: Customer table.
2. Occupancy Tracking
 - Description: A report listing rooms that are presently inhabited or booked for the current date.
 - Information: RoomNumber, RoomType, RoomStatus.
 - Source: (via Room → Reserves → Reservation).
3. Daily Arrival Schedule
 - Description: Displays a list of guests scheduled to check into the hotel on a particular date.
 - Information: FullName, RoomNumber, CheckInDate.
 - Source: (via Customer → Reserves → Reservation).
4. Workforce Allocation
 - Description: Illustrates the staff member assigned to manage or maintain each specific room.
 - Information: RoomNumber, StaffFullName, Role.
 - Source: (via Staff → Manages → Room).
5. Transaction Statement
 - Description: Presents a detailed history of payments finalized by a specific guest.

- Information: TransactionRef, Amount, PaymentStatus.
- Source: (via Customer → Makes → Payment).

Global enhanced entity relationship diagram (EER):

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Relational Schema:

Customer:(IDNumber, FullName, PhoneNumber, EmailAddress, Nationality)

Primary Key: IDNumber

Room: (RoomNumber, RoomType, FloorNumber, Capacity, PricePerNight, RoomStatus)

Primary Key: RoomNumber

Staff:(StaffID, FullName, Role, PhoneNumber, Email)

Primary Key: StaffID

Payment: (TransactionRef, PaymentDate, Amount, PaymentMethod, PaymentStatus, IDNumber)

Primary Key: TransactionRef

Foreign Key: IDNumber references Customer(IDNumber)

Reservation: (IDNumber, RoomNumber, CheckInDate, CheckOutDate, NumberOfGuests, ReservationStatus, BookingDate, TransactionRef)

Primary Key: (IDNumber, RoomNumber)

Foreign Key:

IDNumber references Customer(IDNumber)

RoomNumber references Room(RoomNumber)

TransactionRef references Payment(TransactionRef)

Manages: (StaffID, RoomNumber)

Primary Key: StaffID, RoomNumber

Foreign Key: StaffID references Staff(StaffID)

Foreign Key: RoomNumber references Room(RoomNumber)

Data Dictionary showing description of all entities

Entity Name	Description	Occurrence
Customer	Customers who interact with the hotel reservation system by making room reservations, including personal and contact information such as ID number, full name, phone number, email address, and nationality.	- A customer can make one or more reservations. - A customer may have zero or one payment.
Reservation	Reservations made by customers for hotel rooms, including details such as check-in date, check-out date, number of guests, reservation status, and booking date.	- One reservation is made by one customer. - One reservation is associated with one room. - A reservation may have zero or one payment.
Room	A hotel room available for reservation, including details such as room number, room type, floor number, capacity, price per night, and room status.	- A room can be reserved zero or more times over time. - A room is managed by one or more staff members.
Payment	Payment transactions related to reservations, including transaction reference, payment date, amount, payment method, and payment status.	- One payment is made for one reservation only. - A reservation may have zero or one payment.
Staff	Hotel staff members responsible for managing rooms and handling hotel operations, including staff ID, full name, role, phone number, and email.	- A staff member can manage one or more rooms. - A room can be managed by one or more staff members.

Data Dictionary showing description of all relationships

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Customer	1..*	Reserves	Room	0..*
Customer	1..*	Makes	Payment	0..1
staff	1..*	Manages	Room	1..*

Data Dictionary showing description of all entities

Entity Name	Attribute	Description	Data Type	Length	Nulls	Multi-Valued	Default Value	Range	PK
Customer	IDNumber	A unique identifier for each customer	VarChar	10	No	No			yes
	FullName	The customer's full name.	VarChar	50	No	No			No
	PhoneNumber	The customer's phone number for contact	VarChar	15	No	No			No
	EmailAddress	The customer's email address.	VarChar	50	yes	No			No
	Nationality	The customer's nationality.	VarChar	30	yes	No			No
Reserves	CheckInDate	The date of the guest's check-in.	Date		No	No			No

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	CheckOutDate	The date of the guest's check-out.	Date		No	No			No
	NumberOfGuests	The number of guests included in the reservation.	INT	2	No	No			No
	ReservationStatus	The status of the reservation (confirmed, pending, cancelled).	VarChar	20	No	No			No
	BookingDate	The date on which the reservation was made.	Date		No	No			No
Room	RoomNumber	A unique number to identify each room.	VarChar	10	No	No			yes
	RoomType	The type of room (single, double, suite).	VarChar	20	No	No			No
	FloorNumber	The floor on which the room is located.	INT	2	No	No			No
	Capacity	The maximum number of guests allowed in the room.	INT	2	No	No			No
	PricePerNight	The room price per night.	Decimal	10, 2	No	No			No
	RoomStatus	The current room status (available, occupied,	VarChar	20	No	No			No

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		under maintenance).							
Payment	TransactionRef	A unique reference number for each payment transaction.	VarChar	12	No	No			yes
	PaymentDate	The date on which the payment was made.	Date		No	No			No
	Amount	The amount paid for the reservation.	Decimal	10, 2	No	No			No
	PaymentMethod	The method used to make the payment (., cash, credit card).	VarChar	20	No	No			No
	PaymentStatus	The current payment status (., paid, unpaid, refunded).	VarChar	20	No	No			No
Staff	StaffID	A unique number to identify each staff member.	VarChar	10	No	No			yes
	FullName	The full name of the staff member.	VarChar	50	No	No			No
	Role	The job role of the staff member (e.g., receptionist, manager).	VarChar	30	No	No			No
	PhoneNumber	The staff member's phone number.	VarChar	15	yes	No			No
	Email	The staff member's email address.	VarChar	50	yes	No			No

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Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
IDNUMBER	VARCHAR2(10 BYTE)	N		1		
FULLNAME	VARCHAR2(50 BYTE)	N				
PHONENUMBER	VARCHAR2(15 BYTE)	N				
EMAILADDRESS	VARCHAR2(50 BYTE)	Y				
NATIONALITY	VARCHAR2(30 BYTE)	Y				

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CREATE TABLE "CUSTOMER"

(

"IDNUMBER" VARCHAR2(10),

"FULLNAME" VARCHAR2(50) NOT NULL ENABLE,

"PHONENUMBER" VARCHAR2(15) NOT NULL ENABLE,

"EMAILADDRESS" VARCHAR2(50),

"NATIONALITY" VARCHAR2(30),

PRIMARY KEY ("IDNUMBER")

USING INDEX ENABLE

) ;

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Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
STAFFID	VARCHAR2(10 BYTE)	N		1		
ROOMNUMBER	VARCHAR2(10 BYTE)	N		2		

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CREATE TABLE "MANAGES"

2

(

3

"STAFFID" VARCHAR2(10),

4

"ROOMNUMBER" VARCHAR2(10),

5

PRIMARY KEY ("STAFFID", "ROOMNUMBER")

6

USING INDEX ENABLE

7

);

8

ALTER TABLE "MANAGES" ADD FOREIGN KEY ("STAFFID")

9

REFERENCES "STAFF" ("STAFFID") ENABLE;

10

ALTER TABLE "MANAGES" ADD FOREIGN KEY ("ROOMNUMBER")

11

REFERENCES "ROOM" ("ROOMNUMBER") ENABLE;

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Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
TRANSACTIONREF	VARCHAR2(12 BYTE)	N		1		
PAYMENTDATE	DATE	Y				
AMOUNT	NUMBER(10,2)	Y				
PAYMENTMETHOD	VARCHAR2(20 BYTE)	Y				
PAYMENTSTATUS	VARCHAR2(20 BYTE)	Y				
IDNUMBER	VARCHAR2(10 BYTE)	Y				

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```
CREATE TABLE "PAYMENT"
(
  "TRANSACTIONREF" VARCHAR2(12),
  "PAYMENTDATE" DATE,
  "AMOUNT" NUMBER(10,2),
  "PAYMENTMETHOD" VARCHAR2(20),
  "PAYMENTSTATUS" VARCHAR2(20),
  "IDNUMBER" VARCHAR2(10),
  PRIMARY KEY ("TRANSACTIONREF")
  USING INDEX ENABLE
);

ALTER TABLE "PAYMENT" ADD FOREIGN KEY ("IDNUMBER")
REFERENCES "CUSTOMER" ("IDNUMBER") ENABLE;
```

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Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
IDNUMBER	VARCHAR2(10 BYTE)	N		1		
ROOMNUMBER	VARCHAR2(10 BYTE)	N		2		
CHECKINDATE	DATE	Y				
CHECKOUTDATE	DATE	Y				
NUMBEROFGUESTS	NUMBER(2,0)	Y				
RESERVATIONSTATUS	VARCHAR2(20 BYTE)	Y				
BOOKINGDATE	DATE	Y				
TRANSACTIONREF	VARCHAR2(12 BYTE)	Y				

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```
CREATE TABLE "RESERVATION"
(
  "IDNUMBER" VARCHAR2(10),
  "ROOMNUMBER" VARCHAR2(10),
  "CHECKINDATE" DATE,
  "CHECKOUTDATE" DATE,
  "NUMBEROFGUESTS" NUMBER(2,0),
  "RESERVATIONSTATUS" VARCHAR2(20),
  "BOOKINGDATE" DATE,
  "TRANSACTIONREF" VARCHAR2(12),
  PRIMARY KEY ("IDNUMBER", "ROOMNUMBER")
  USING INDEX ENABLE
);

ALTER TABLE "RESERVATION" ADD FOREIGN KEY ("IDNUMBER")
REFERENCES "CUSTOMER" ("IDNUMBER") ENABLE;
ALTER TABLE "RESERVATION" ADD FOREIGN KEY ("ROOMNUMBER")
REFERENCES "ROOM" ("ROOMNUMBER") ENABLE;
ALTER TABLE "RESERVATION" ADD FOREIGN KEY ("TRANSACTIONREF")
REFERENCES "PAYMENT" ("TRANSACTIONREF") ENABLE;
```

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Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
ROOMNUMBER	VARCHAR2(10 BYTE)	N		1		
ROOMTYPE	VARCHAR2(20 BYTE)	N				
FLOORNUMBER	NUMBER(2,0)	N				
CAPACITY	NUMBER(2,0)	N				
PRICEPERNIGHT	NUMBER(10,2)	N				
ROOMSTATUS	VARCHAR2(20 BYTE)	N				

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```
1 CREATE TABLE "ROOM"
2 (
3   "ROOMNUMBER" VARCHAR2(10),
4   "ROOMTYPE" VARCHAR2(20) NOT NULL ENABLE,
5   "FLOORNUMBER" NUMBER(2,0) NOT NULL ENABLE,
6   "CAPACITY" NUMBER(2,0) NOT NULL ENABLE,
7   "PRICEPERNIGHT" NUMBER(10,2) NOT NULL ENABLE,
8   "ROOMSTATUS" VARCHAR2(20) NOT NULL ENABLE,
9   PRIMARY KEY ("ROOMNUMBER")
10  USING INDEX ENABLE
11 ) ;
```

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Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
STAFFID	VARCHAR2(10 BYTE)	N		1		
FULLNAME	VARCHAR2(50 BYTE)	N				
ROLE	VARCHAR2(30 BYTE)	N				
PHONENUMBER	VARCHAR2(15 BYTE)	Y				
EMAIL	VARCHAR2(50 BYTE)	Y				

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CREATE TABLE "STAFF"

(

"STAFFID" VARCHAR2(10),

"FULLNAME" VARCHAR2(50) NOT NULL ENABLE,

"ROLE" VARCHAR2(30) NOT NULL ENABLE,

"PHONENUMBER" VARCHAR2(15),

"EMAIL" VARCHAR2(50),

PRIMARY KEY ("STAFFID")

USING INDEX ENABLE

) ;

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	IDNUMBER	FULLNAME	PHONENUMBER	EMAILADDRESS	NATIONALITY
✎	C002	Noura	0552222222	noura@gmail.com	Saudi
✎	C001	Sara	0551111111	sara@gmail.com	Saudi

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	STAFFID	ROOMNUMBER
✎	S001	R101
✎	S002	R102

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+ Insert RowColumns...Filter...Count RowsLoad DataDownloadRefresh

	TRANSACTIONREF	PAYMENTDATE	AMOUNT	PAYMENTMETHOD	PAYMENTSTATUS	IDNUMBER
✎	T002	4/29/2026	500	Card	Paid	C002
✎	T001	4/29/2026	300	Cash	Paid	C001

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	IDNUMBER	ROOMNUMBER	CHECKINDATE	CHECKOUTDATE	NUMBEROFGUESTS..	RESERVATIONSTATU\$	BOOKINGDATE	TRANSACTIONREF
✎	C001	R101	4/29/2026	5/1/2026	1	Confirmed	4/29/2026	T001
✎	C002	R102	4/29/2026	5/2/2026	2	Confirmed	4/29/2026	T002

1 cells selected

1 - 2

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Data insertion:

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+ Insert RowColumns...Filter...Count RowsLoad DataDownloadRefresh

	ROOMNUMBER	ROOMTYPE	FLOORNUMBER	CAPACITY	PRICEPERNIGHT	ROOMSTATUS
	R102	Double	1	2	500	Occupied
	R101	Single	1	1	300	Available

1 cells selected

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ColumnsDataIndexesConstraintsGrantsStatisticsTriggersDependenciesDDLSample Queries

+ Insert RowColumns...Filter...Count RowsLoad DataDownloadRefresh

	STAFFID	FULLNAME	ROLE	PHONENUMBER	EMAIL
	S002	Fahad Saleh	Receptionist	0554444444	fahad@gmail.com
	S001	Ahmed Khalid	Manager	0553333333	ahmed@gmail.com

1 cells selected

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1SELECT * FROM Room
2WHERE RoomStatus = 'Available';

ResultsExplainDescribeSaved SQLHistory

ROOMNUMBER	ROOMTYPE	FLOORNUMBER	CAPACITY	PRICEPERNIGHT	ROOMSTATUS
R101	Single	1	1	300	Available

1 rows returned in 0.01 secondsDownload

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1SELECT s.FullName, m.RoomNumber
2FROM Staff s
3JOIN Manages m ON s.StaffID = m.StaffID;

ResultsExplainDescribeSaved SQLHistory

FULLNAME	ROOMNUMBER
Ahmed Khalid	R101
Fahad Saleh	R102

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1SELECT IDNumber, COUNT(*) AS TotalReservations
2FROM Reservation
3GROUP BY IDNumber;

ResultsExplainDescribeSaved SQLHistory

IDNUMBER	TOTALRESERVATIONS
C001	1
C002	1

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1SELECT c.FullName, r.RoomType, p.Amount

2FROM Reservation res

3JOIN Customer c ON res.IDNumber = c.IDNumber

4JOIN Room r ON res.RoomNumber = r.RoomNumber

5JOIN Payment p ON res.TransactionRef = p.TransactionRef;

Results

Explain

Describe

Saved SQL

History

FULLNAME	ROOMTYPE	AMOUNT
Sara	Single	300
Noura	Double	500

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1SELECT * FROM Payment

2WHERE PaymentStatus = 'Paid';

Results

Explain

Describe

Saved SQL

History

TRANSACTIONREF	PAYMENTDATE	AMOUNT	PAYMENTMETHOD	PAYMENTSTATUS	IDNUMBER
T002	4/29/2026	500	Card	Paid	C002
T001	4/29/2026	300	Cash	Paid	C001

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1SELECT c.FullName, r.RoomNumber, res.CheckInDate

2FROM Reservation res

3JOIN Customer c ON res.IDNumber = c.IDNumber

4JOIN Room r ON res.RoomNumber = r.RoomNumber;

Results

Explain

Describe

Saved SQL

History

FULLNAME	ROOMNUMBER	CHECKINDATE
Sara	R101	4/29/2026
Noura	R102	4/29/2026

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1SELECTAVG(PricePerNight)ASAveragePrice

2FROMRoom;

Results

Explain

Describe

Saved SQL

History

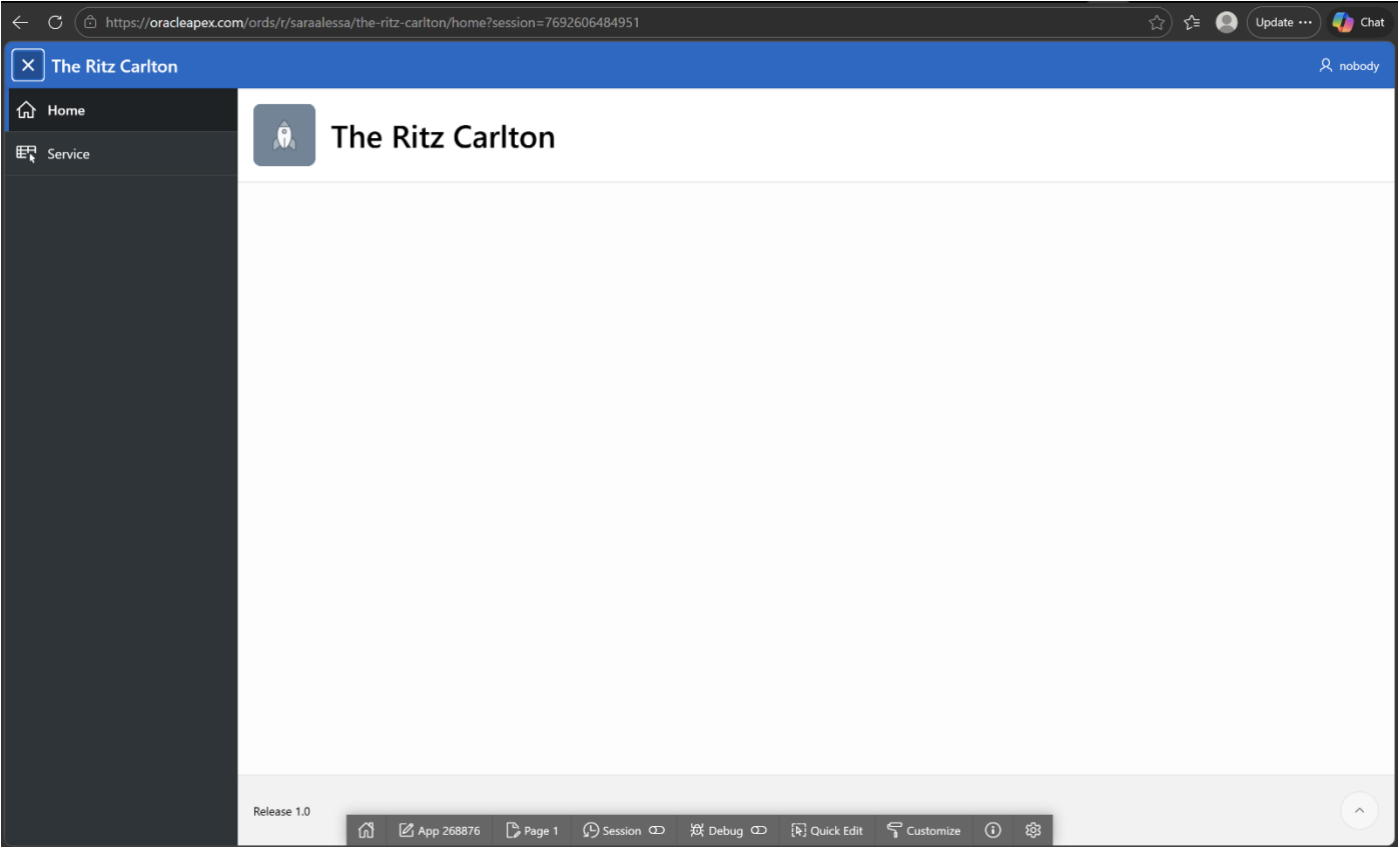
AVERAGEPRICE

400

1 rows returned in 0.01 seconds

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App Builder:



App Builder:

- Interactive Grid:

The screenshot displays the Oracle APEX application 'The Ritz Carlton' in a web browser. The application has a blue header bar with the title 'The Ritz Carlton' and a user profile 'nobody'. A dark sidebar on the left contains navigation links for 'Home' and 'Service'. The main content area is titled 'Service' and features an Interactive Grid. The grid has two columns: 'Name' and 'Price'. A single row is visible in the grid, highlighted in light green, with a checkmark in the selection column. Above the grid, there is a search bar with the text 'Search: All Text Columns', a 'Go' button, and action buttons for 'Actions', 'Edit', 'Save', and 'Add Row'. A 'Reset' button is also present. Below the grid, it indicates '1 rows selected' and 'Total 1'. The bottom of the screen shows a status bar with 'Release 1.0' and various application controls like 'App 268876', 'Page 2', 'Session', 'Debug', 'Quick Edit', and 'Customize'.

	Name	Price
☒		

1 rows selected Total 1

App Builder:

- Master Detail:

The Ritz Carlton

Home

Service

Customer Reservation

Customer Reservation

Save

Search: All Text ColumnsGoActionsEditAdd RowReset

	Fullname	Phonenumber	Emailaddress	Nationality
☒	Noura	0552222222	noura@gmail.com	Saudi
☐	Sara	0551111111	sara@gmail.com	Saudi

1 rows selectedTotal 2

Search: All Text ColumnsGoActionsEditAdd RowReset

	Checkindate	Checkoutdate	Numberofguests	Reservationstatus	Bookingdate	Transactionref
☒	4/29/2026	5/2/2026	2	Confirmed	4/29/2026	T002

1 rows selectedTotal 1

Release 1.0

App 268876Page 4SessionDebugQuick EditCustomize

